Spatial Context Causal Adjustment: An Open Diagnostic Workflow for Geographic Observational Studies

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Abstract

Geographic observational studies often combine spatially structured treatments, outcomes, and environmental context. Standard causal estimators can be used in these studies, but their credibility depends on whether spatial context is a defensible adjustment source and on whether residual spatial structure is reported rather than hidden.

We introduce Spatial Context Causal Adjustment (SCCA), an open audit workflow for constructing and diagnosing spatial-context adjustment sets. SCCA is not a new identification theorem. It standardises candidate context construction, adjustment-set comparison, common-support checks, post-adjustment balance, spatial block bootstrap, threshold placebo tests, residual spatial autocorrelation, spatial-lag and graph sensitivity, variable-role audits, and rule-based evidence grades. Residual Moran's I is treated as a downgrade diagnostic, not as an identifying statistic. Its default material threshold is calibrated in a 2,880-fit controlled design spanning SAR, CAR, distance-kernel, and non-stationary latent spatial processes.

We evaluate SCCA with synthetic stress tests, a Chongqing urban heat case, a county GIS reproducibility check, and a synthetic positive control. In Chongqing, the treatment is building-level high-rise status but the outcome is MODIS summer LST on a $\sim$1 km grid. The primary empirical quantity is therefore an outcome-scale pixel model: the slope relating pixel-mean high-rise share to pixel-mean LST was 0.546°C per unit share (95% CI [0.359, 0.734]), not a building-level ATT. The building-level matching ATT was 0.303°C (95% CI [0.220, 0.383]) and is retained only as a diagnostic approximation. Clustering on the shared outcome pixel widened the building-level OLS standard error by 41.5%. The preferred pre-treatment matching design slightly missed the strict balance rule (maximum post-match SMD = 0.104), and the matched residual Moran's $I = 0.112$ ($p = 0.01$) crossed the default 0.10 material threshold by 0.012. The grade is therefore bounded support under the default rule and remains a residual-spatial warning under higher thresholds. The calibration also shows that this warning is process-dependent: at $t = 0.10$, true-positive rates are 0.194 for CAR, 0.529 for SAR, 0.994 for distance-kernel, and 1.000 for non-stationary latent fields. The county dataset is retained as an ArcGIS-free workflow and spatial-diagnostic check rather than a substantive validation case, and a synthetic positive control is graded core support.

SCCA contributes a reproducible protocol for auditing evidence in geographic causal analysis. It does not remove unmeasured spatial confounding, solve interference, or make restricted data fully reproducible; it makes adjustment choices, spatial diagnostics, and claim boundaries inspectable across GIS-facing workflows.

Keywords: spatial causal inference; spatial confounding; causal adjustment; evidence grading; modifiable areal unit problem; open reproducibility

1 Introduction

Geographic information science increasingly supports causal policy questions: whether urban form changes heat exposure, whether infrastructure changes disease risk, or whether land-use interventions change environmental outcomes. These questions are usually studied from observational spatial data, where treatment is rarely randomized, units are embedded in environmental gradients, and outcomes may be measured at resolutions that differ from the intervention units.

The main challenge is not the absence of causal estimators. Matching, weighting, regression adjustment, difference-in-differences, spatial panel models, and machine-learning estimators are available to GIS researchers [Stuart, 2010, Angrist and Pischke, 2009, Athey and Imbens, 2019, Hernán and Robins, 2020]. The harder question is how to use spatial context. Elevation, land cover, access, morphology, settlement history, and administrative structure can confound geographic effects, while spatial data also introduce dependence, possible interference, and spatially structured unmeasured confounding [Reich et al., 2020, Gilbert et al., 2021, Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023]. Omitting context leaves confounding; adding it indiscriminately can reduce support or adjust for invalid post-treatment variables.

Modern GIS workflows intensify this problem because remote-sensing indices, terrain, land cover, geometry, and neighborhood descriptors are easy to extract at scale [Anselin and Rey, 2014, Brunsdon, 2016]. Recent spatial-confounding work also shows that spatial smooths or fixed effects can corrupt the treatment coefficient when treatment is spatially patterned [Hodges and Reich, 2010, Dupont et al., 2022, Tec et al., 2023]. Geographic causal analysis therefore needs a workflow that asks which context variables improve credible adjustment, under visible diagnostics, and where the evidence stops.

This paper proposes Spatial Context Causal Adjustment (SCCA), a workflow rather than a new estimator or identification theorem. SCCA wraps standard estimators with context construction, adjustment-set comparison, common-support and balance diagnostics, spatial robustness tests, and an evidence grade. Its conservative premise is that context supports causal adjustment only when it improves a diagnosed design, and a positive estimate should be downgraded when spatial diagnostics show remaining structure.

Our contributions are:

1. A formal SCCA workflow for selecting and diagnosing spatial-context adjustment sets, positioned as an audit protocol rather than a universal estimator.

2. A residual-spatial downgrade rule calibrated across SAR, CAR, kernel, and non-stationary latent processes, reporting where residual Moran's I transfers and where it does not.

3. A reproducible evidence protocol covering support, balance, spatial bootstrap, threshold placebos, residual autocorrelation, neighborhood sensitivity, variable-role auditing, change-of-support diagnostics, and evidence grading.

4. Focused evidence from synthetic stress tests, a Chongqing urban heat case, a county notebook/GIS run, and a synthetic positive control certified as core support.

Chongqing is the main case because author-controlled spatial processing links candidate context variables directly to the adjustment design. The county case is more open but uses third-party training/demo data, so it is used for ArcGIS-free packaging, spatial downgrading, and interface reproducibility rather than for the main causal claim.

The paper makes a bounded claim. SCCA does not eliminate unobserved spatial confounding, solve spatial interference, or prove that richer adjustment is always better. It makes such claims harder to make without diagnostic evidence.

2 Related Work

2.1 Causal inference with spatial data

Potential-outcome causal inference compares outcomes under treatment and control states [Imbens and Rubin, 2015, Hernán and Robins, 2020]. Observational designs require conditional exchangeability, often operationalised by matching or weighting [Rosenbaum and Rubin, 1983, Stuart, 2010]. Spatial studies make this difficult because location and environment are structured confounders. Directed acyclic graphs help separate confounders, mediators, colliders and post-treatment descendants [Pearl, 2009, VanderWeele, 2015]; we use this framing for candidate spatial context (Figure 1).

Spatial econometrics and spatial statistics provide tools for autocorrelation, spillover, and dependence [Anselin, 1988, LeSage and Pace, 2009, Anselin and Rey, 2014]. Spatial causal inference has addressed unmeasured spatial confounding, interference, and neighborhood exposure [Reich et al., 2020, Gilbert et al., 2021, Giffin et al., 2020, Papadogeorgou et al., 2022, Schnell and Papadogeorgou, 2020]. A key warning is that adding spatial smooths or fixed effects can bias a spatially patterned treatment coefficient [Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023]; spatial double-machine-learning work similarly requires geographic cross-fitting blocks [Tec et al., 2023]. These tools do not by themselves select valid adjustment sets for routine GIS workflows. SCCA therefore treats spatial-dependence diagnostics and causal-adjustment diagnostics as related but distinct.

2.2 Spatial context as an adjustment source

Geographic adjustment sets often include coordinates, fixed effects, distance, terrain, land cover, built environment, and remote-sensing indices. These variables can proxy for unmeasured context, but only under assumptions about timing, causal role, and compatible spatial scale [Gilbert et al., 2021, VanderWeele, 2015]. They can also be collinear, weakly related to treatment, scale-mismatched, or post-treatment. Because scale and aggregation interact with MAUP and edge effects [Openshaw, 1984, Fotheringham and Wong, 1991, Manley et al., 2006], SCCA requires support and balance checks before context supports a causal statement, and reports graph and threshold sensitivity.

2.3 Continuous exposure, spatial interference, and diagnostics

Many GIS applications use continuous or ordered exposures. Generalised propensity-score methods estimate dose-response functions [Hirano and Imbens, 2004, Zhao et al., 2013]. Spatial settings also raise interference concerns because nearby exposure can affect outcomes. Formal spatial-interference methods address this in specialised designs [Verbitsky-Savitz and Raudenbush, 2012, Aronow and Samii, 2017, Forastiere et al., 2021, Giffin et al., 2020, Papadogeorgou et al., 2022]. SCCA does not identify spillovers in that sense; it reports neighborhood-exposure and spatial-lag outputs as stability diagnostics.

2.4 Diagnostics and evidence grades

Causal estimates are often reported as single numbers even when the adjustment design is fragile. SCCA instead interprets effects together with balance, matched counts, support attrition, bootstrap stability, placebo checks, residual autocorrelation, neighbor-exposure sensitivity, and graph-specification sensitivity. The evidence grade is not a significance label; it records whether the declared rule table gives core or bounded support.

2.5 Position relative to adjacent workflows

SCCA is an audit layer connecting causal adjustment diagnostics with GIS-facing spatial diagnostics. It reuses standard matching, weighting, regression, GPS and SLX-style sensitivity models, but adds a shared request schema, alternative context variants, support and balance checks, residual-spatial downgrades, and portable GIS outputs. Relative to spatial-confounding work [Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023, Gilbert et al., 2021, Tec et al., 2023], the increment is operational: select the reported specification by declared causal role, calibrate residual-spatial downgrades, and carry change-of-support diagnostics into one portable evidence contract. This reproducible audit layer is consistent with recent GIScience work on reusable computational geography [Nüst and Pebesma, 2021, Konkol and Kray, 2019, Brunsdon, 2016].

3 Spatial Context Causal Adjustment

3.1 Problem formulation

Let $i = 1, \ldots, n$ index geographic units with geometry g_i, treatment or exposure T_i, outcome Y_i, observed non-spatial covariates X_i, and candidate spatial-context variables C_i. The context vector may contain coordinates, geometry, terrain, land cover, remote-sensing bands or indices, and neighborhood summaries. Figure 1 represents the Chongqing case: observed terrain and historical land use are candidate confounders, latent U is unmeasured spatial structure, and NDVI is shown as a possible mediator requiring a role decision.

For binary treatment, the target estimand in the main case study is the average treatment effect on the treated:

$$\tau_{\mathrm{ATT}} = E[Y_i(1) - Y_i(0) \mid T_i = 1]. \tag{1}$$

This estimand is meaningful when treatment and outcome share a unit. When the outcome support is coarser, SCCA reports an outcome-scale estimand. In Chongqing, buildings are aggregated to MODIS LST pixels p, with pixel-mean high-rise share $\bar{T}_p$ and pixel-mean LST $\bar{Y}_p$. The primary coefficient is

$$\beta_{\mathrm{pixel}} = \frac{\partial E[\bar{Y}_p \mid \bar{T}_p, \bar{X}_p, \bar{C}_p]}{\partial \bar{T}_p}, \tag{2}$$

estimated by an adjusted pixel-level outcome model. It is a high-rise-share slope at the outcome support, not a building-level ATT. Other continuous exposures use exposure-response functions or local slopes with the same diagnostics.

SCCA does not identify τ_{ATT} or β_{pixel} by itself. Identification still requires measured confounders, overlap, and limited or modeled interference. SCCA tests whether a chosen context set can support a bounded causal interpretation.

3.2 Identification assumptions and scope

The workflow rests on standard observational-causal assumptions, stated for spatial data [Imbens and Rubin, 2015, Pearl, 2009]. Consistency requires the

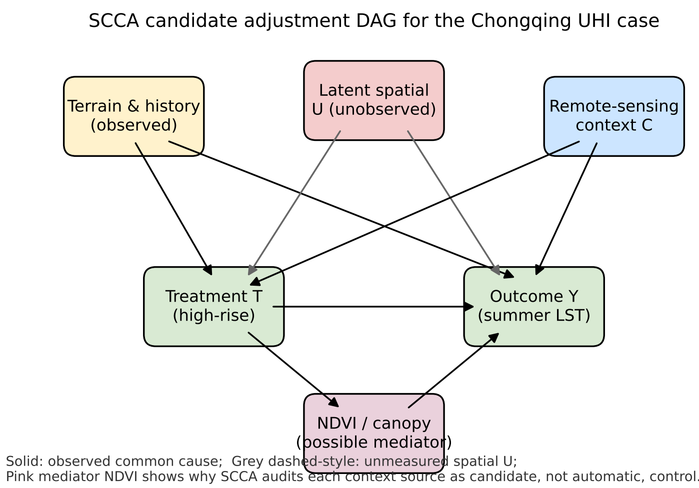

Figure 1: Candidate adjustment DAG for the Chongqing UHI case. Terrain and historical land-use are observed common causes of treatment T (high-rise) and outcome Y (summer LST). A latent spatial process U remains unobserved; SCCA reports residual Moran's I and spatial-lag sensitivity as diagnostics for the resulting bias rather than identifying U. NDVI is shown explicitly as a possible post-treatment mediator: SCCA's variable-role layer requires the analyst to declare whether it is added as confounder or omitted as descendant of T.

exposure, outcome, and spatial unit to match the intervention. Conditional exchangeability requires observed covariates and context to contain relevant common causes; coordinates and remote-sensing variables are proxies, not guarantees [Gilbert et al., 2021]. Positivity requires overlap, so SCCA reports support loss and balance. No-interference must be plausible or bounded, and scale compatibility is required when exposure, context, and outcome have different resolutions. The evidence grade labels claim strength under these assumptions.

3.3 Workflow

Figure 2 summarizes the workflow: construct candidate context features, define adjustment variants, estimate the appropriate treatment or outcome model, diagnose support and balance, then run robustness checks and assign an evidence grade.

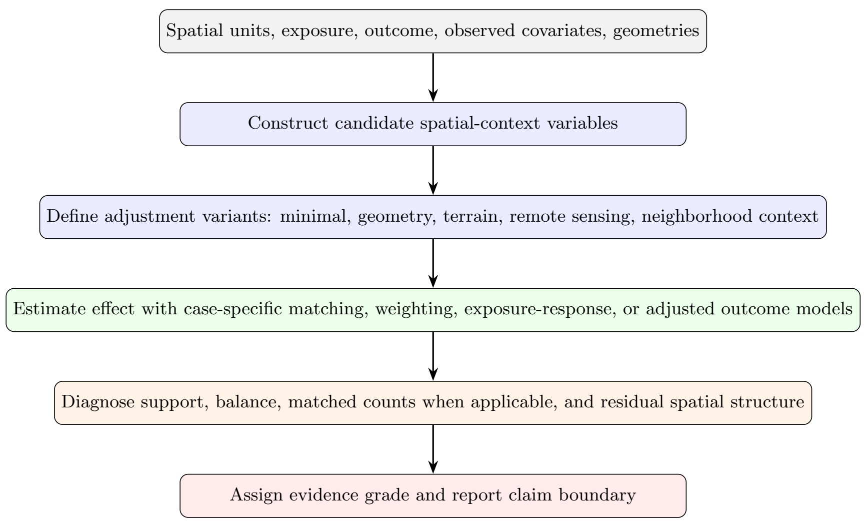

Figure 2: SCCA workflow. Spatial context is treated as a candidate adjustment source and must pass support, balance, and robustness diagnostics before it is used to support a causal claim.

The estimators are deliberately standard. The contribution is the diagnosable adjustment workflow and portable evidence contract.

3.4 Adjustment variants

SCCA does not impose one estimator on all exposure types. The reusable core estimates adjusted outcome models and ERFs for continuous or ordered exposures using generalized propensity-score weights [Hirano and Imbens, 2004, Zhao et al., 2013]. Case modules can add binary-treatment estimators. In

Chongqing, the binary module estimates

$$e_i = P(T_i = 1 \mid X_i, C_i) \tag{3}$$

for each adjustment variant. Units outside overlap are removed, treated units are nearest-neighbor matched within caliper, and standardized mean differences (SMDs) are computed:

$$\mathrm{SMD}_k = \frac{\bar{Z}_{k,T=1} - \bar{Z}_{k,T=0}}{\sqrt{(s^2_{k,T=1} + s^2_{k,T=0})/2}}, \tag{4}$$

where Z_k is the kth covariate. A variant is credible only if the maximum post-match absolute SMD is below 0.1 unless justified, and estimates are reported with matched counts and support attrition.

For continuous exposures, SCCA reports adjusted OLS, difference-outcome OLS when a baseline outcome exists, and a GPS-weighted exposure-response function. Model-based intervals describe fitted-outcome uncertainty; the evidence grade also uses spatial bootstrap, placebo, support, and residual-spatial diagnostics.

3.5 Robustness diagnostics

When data permit, SCCA runs spatial block bootstrap, placebo exposure definitions, weaker-outcome or placebo-outcome checks, and residual spatial autocorrelation diagnostics.

For spatial datasets, SCCA builds a neighborhood graph from polygon adjacency or coordinate k-nearest neighbors. The graph supports Moran diagnostics, neighbor-exposure adjustment, graph sensitivity, and an SLX-style sensitivity model [LeSage and Pace, 2009]:

$$Y_i = \alpha + \beta T_i + \theta (WT)_i + \gamma X_i + \phi C_i + \eta (WX)_i + \lambda (WC)_i + \varepsilon_i, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \tag{5}$$

where W is row-standardised. We interpret β as the local adjusted association and θ as a neighboring-exposure signal. This is a stability diagnostic, not a full interference design.

3.6 Diagnostic interpretation of residual Moran's I

The residual-Moran rule is a warning boundary, not an identification result. Consider

$$Y_i = \tau T_i + X_i^\top \gamma + C_i^\top \phi + \sigma_u U_i + \nu_i, \tag{6}$$

where $U = (I - \rho W)^{-1}\epsilon$ is a stationary SAR(ρ) field and ν_i is i.i.d. noise. If SCCA adjusts for (X, C) but not U, the fitted residual can be written heuristically as

$$r \approx M\nu + \sigma_u M U, \tag{7}$$

where M removes the fitted adjustment design. Residual Moran's I can therefore increase when omitted spatial structure remains, but it does not identify ρ, σ_u, or treatment bias, and a non-significant value does not rule out spatial confounding.

SCCA therefore asks a narrower question: after the declared adjustment set is fitted, is residual spatial structure large enough to make a stronger causal statement unsafe under the rule table? Sections 4.8 and 4.9 make this threshold auditable while keeping it diagnostic.

3.7 Evidence grades and threshold calibration

SCCA assigns evidence grades with a machine-readable rule file. *Core support* requires strong credibility, robust support, and no material spatial downgrade. *Bounded support* marks useful evidence with explicit support, scale, balance, robustness, residual-spatial, or reproducibility limits.

The manuscript uses the 2026-06-30 evidence-grade rule set. Downgrades are triggered by moderate or weak credibility, weak support or robustness, large exposure-balance or boundary diagnostics, material residual Moran's I, significant neighbor exposure, large spatial-adjustment shifts, or graph-sensitivity sign changes. The default residual-Moran threshold is $t_{\mathrm{Moran}} = 0.10$; the full table is exported as `07_results/scca_evidence_grade_rules.json` and `07_results/scca_evidence_grade_rules.md`.

These thresholds are audit rules, not validity constants. The SMD threshold 0.10 follows matching practice [Stuart, 2010, Austin, 2009]; other non-Moran cutoffs remain conservative manuscript defaults. Only the residual-Moran threshold is calibrated here by sweeping spatial-process family, ρ, n and σ_u/σ_ν and comparing true-positive and false-positive rates. The default 0.10 is a low-false-positive warning boundary; 0.20 is retained as a high-specificity check.

A downgrade does not prove bias, and its absence does not prove identification. It only records whether a declared warning boundary was crossed. Statistically significant but sub-threshold residual Moran's I is also recorded in `07_results/scca_grade_threshold_sensitivity.csv`.

3.8 Implementation surfaces

SCCA is implemented as a shared Python core with thin CLI, notebook, ArcGIS Pro, and QGIS Processing adapters. The reusable `AnalysisRequest` records inputs, fields, covariates, context columns, coordinates, bootstrap groups, placebo settings, trimming thresholds, target outcomes, and *variable-role* declarations. Runs write CSV, JSON, Markdown diagnostics and, for spatial inputs, common GIS outputs and QGIS `.qml` styles.

The binary Chongqing module separately writes ablation, balance, matched-count, change-of-support, and matching-sensitivity tables. These outputs distinguish the outcome-scale pixel estimand from building-level matching diagnostics.

3.9 Missing data, scale, and uncertainty

Three conservative choices are recorded explicitly. Missing context values use within-variant median imputation with a missingness indicator added to X; spatially structured missingness should also trigger complete-case sensitivity. Scale and MAUP [Openshaw, 1984, Fotheringham and Wong, 1991, Manley et al., 2006] enter through the treatment unit, context resolution, and diagnostic graph. When the outcome is coarser than treatment, SCCA treats the outcome-scale coefficient as primary. In Chongqing, this is β_{pixel}, the pixel-level slope relating mean high-rise share to mean LST; building-level ATT and clustered OLS remain diagnostics. SCCA also reports graph-sensitivity across $k \in \{4, 6, 8, 12\}$ and threshold-placebo sensitivity, but it does not solve scale mismatch. Uncertainty intervals come from HC3 or clustered standard errors, spatial block bootstrap quantiles, or Abadie–Imbens matching standard errors [Abadie and Imbens, 2006], as indicated in each table.

4 Experiments

4.1 Dataset roles and case selection

The experiments assign different evidentiary roles to the datasets. Chongqing is the main real-data SCCA case because its high-rise treatment, geometry, terrain, remote-sensing context, and summer LST outcome are tightly coupled. It tests whether richer spatial context can support adjustment without overstating the association.

The county social-capital data instead test ArcGIS-free rerun counts, output contracts, residual spatial diagnostics, and downgrading behavior for a familiar GIS example. The reproducibility scope differs by dataset (Table 1); raw building-level Chongqing inputs are not redistributed.

Table 1: Reproducibility scope of the experimental cases.

Case	Public rerun level	Restricted or source-governed inputs	Manuscript use
Synthetic benchmark audit	Exact public rerun from committed generators and scripts	None beyond software dependencies	Stress-test estimator behavior under known targets
Chongqing urban heat	Structural rerun from public scripts, manifests, and generated summaries	Building footprints, precise coordinates, floor attributes, and derived local environmental variables require approved local access	Main real-data SCCA ablation and claim-boundary example
County social capital GIS run	Exact rerun where users have the licensed third-party example dataset; public outputs and scripts document the contract	Esri training/demo data and source terms govern redistribution and use	ArcGIS-free GIS workflow, cross-interface output, and spatial-diagnostic downgrade check

4.2 Evidence synthesis design

The experiments addressed estimator stress tests, Chongqing diagnostics, county GIS/notebook reproducibility, and whether the grade engine can certify a clean positive control (`07_results/core_support_positive_control.json`).

Table 2 summarizes the evidence matrix. The source table is generated in `07_results/scca_evidence_synthesis.csv`; residual-spatial threshold sensitivity is written to `07_results/scca_grade_threshold_sensitivity.csv`.

Table 2: SCCA evidence synthesis. Grades describe claim strength, not statistical significance alone.

Case	Data type	Key diagnostic	Grade	Manuscript use
Synthetic benchmark audit	Controlled generators	48 benchmark rows; 13 robust and 29 fragile rows	Bounded support	Estimator stress test, not real-world validation
Chongqing urban heat	Real building and remote-sensing data	Pixel high-rise-share slope = 0.546°C per unit share; building-level matching ATT = 0.303 (diagnostic); residual Moran's I = 0.112	Bounded support	Main real-data case with outcome-scale estimand, diagnostic matching, and residual-spatial caution
County social capital spatial notebook	GIS/notebook reproducibility case	residual Moran's I = 0.313; spatial-lag coef. = 0.145; ArcGIS-trimmed ERF range = 6.85 years	Bounded support	Spatial diagnostics and cross-interface output check
Synthetic core-support control	Clean controlled DGP (all confounders measured)	$\widehat{\mathrm{ATT}}$ = 0.505 (true 0.5); residual Moran's $I \approx 0$ (n.s.); no downgrade rule fires	Core support	Positive control showing the core/bounded distinction is discriminative

4.3 Synthetic benchmark audit

The controlled benchmark used six estimator families: matching, DiD, ERF, causal forests, spatial Granger tests, and GCCM. These 48 rows are stress tests, not empirical claims. DiD and causal forests recovered baseline targets, ERF was useful but sometimes biased, and severe-stress settings were fragile. The benchmark supports SCCA as a diagnostic wrapper, not a guarantee that every estimator works in every spatial setting.

4.4 Chongqing urban heat case

The main case asked whether Chongqing high-rise concentration was associated with summer land surface temperature after spatial-context adjustment. The stratified 2021 sample contained 5,000 buildings; treatment was at least 10 floors. MODIS MOD11A2 summer LST (June–August 2021) was sampled on its native ∼1 km grid. Candidate context variables included coordinates, footprint area, Sentinel-2 bands and indices, 30 m SRTM elevation, and slope. Because the outcome is coarser than the treatment unit, the primary quantity is the outcome-scale pixel high-rise-share slope; building-level matching is diagnostic.

Outcome-scale estimate and diagnostic matching. At the outcome resolution, the pre-treatment model estimated a 0.546°C slope (95% CI [0.359, 0.734]) per unit increase in pixel-mean high-rise share (Table 3). This aligns with the MODIS support and is not a building-level ATT. The building-level matching estimate was 0.303°C (95% CI [0.220, 0.383]; Table 4). Clustering adjusted building-level OLS on the shared outcome pixel widened the standard error from 0.038 to 0.054 (41.5%).

Change of support. The 5,000 buildings map to about 1,400 outcome pixels, and buildings within a pixel share the same LST value. Table 3 therefore reports naive building-level OLS, pixel-clustered OLS, and the pixel high-rise-share model. SCCA cannot remove the support mismatch, but it prevents it from being hidden behind one building-level coefficient.

Table 3: Chongqing change-of-support diagnostics for the pre-treatment set. The pixel row is the primary outcome-scale slope; building-level rows are diagnostic.

Estimand	Coefficient	SE	95% CI	Units
Building, naive iid SE	0.268	0.038	[0.193, 0.343]	5,000 buildings
Building, cluster-robust SE	0.268	0.054	[0.162, 0.374]	1,400 pixel clusters
Pixel high-rise-share slope (°C per unit share)	0.546	0.096	[0.359, 0.734]	1,400 pixels

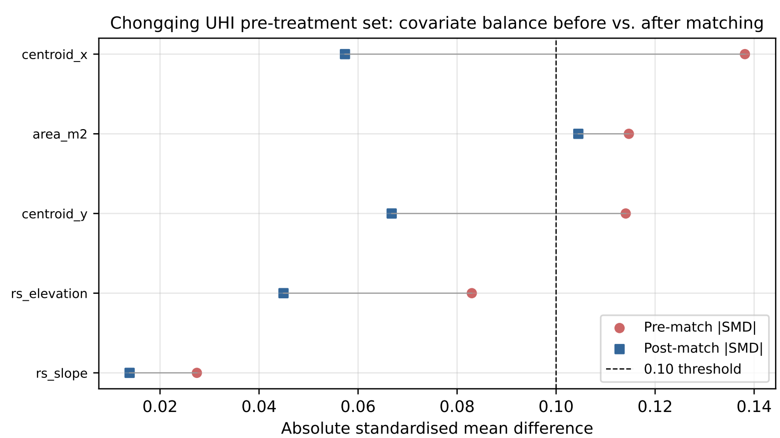

Figure 3: Chongqing UHI covariate balance before and after caliper matching (Love plot) for the pre-treatment set. The 0.10 SMD threshold is shown for reference; the maximum post-match SMD is 0.104, a near-threshold miss rather than a strict balance pass.

The pre-treatment set was chosen for causal role rather than minimum SMD. It uses coordinates, geometry, and terrain, which are more plausibly pre-construction than same-period Sentinel surfaces. Its maximum post-match SMD is 0.104 (`near_threshold_not_passed` in `07_results/chongqing_matching_sensitivity.csv`). The full remote-sensing set improves balance (0.061) and

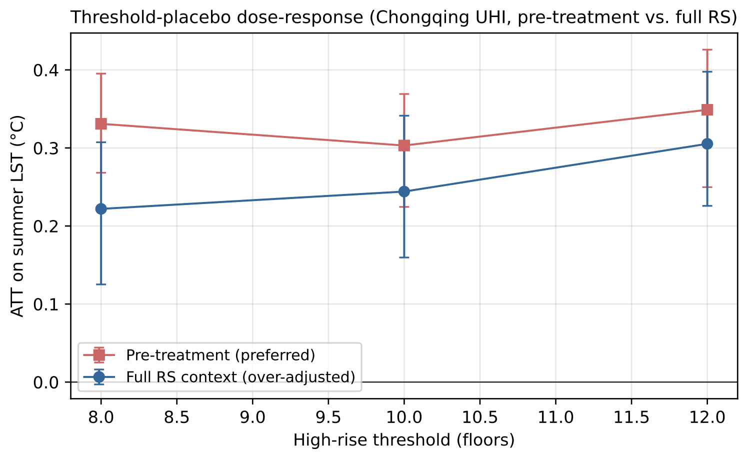

Figure 4: Chongqing threshold-placebo dose-response. Vertical bars are matching-based 95% CIs. The monotone increase from 8 to 12 floors is consistent with a dose-response interpretation; all estimates remain small relative to the thermal-retrieval and change-of-support uncertainties that accompany ∼1 km LST extraction.

Table 4: Chongqing UHI adjustment variants. *pre_treatment* is preferred because it avoids possible post-treatment Sentinel surfaces but slightly misses strict balance. Balance pass means maximum post-match SMD < 0.10.

Variant	ATT (°C)	CI lower	CI upper	Max pre SMD	Max post SMD	Role
Raw difference	0.238	0.163	0.307	–	–	unadjusted
Coordinates only	0.267	0.178	0.346	0.138	0.080	pre-treatment
Geometry	0.252	0.180	0.333	0.138	0.125	pre-treatment
Terrain	0.303	0.229	0.367	0.138	0.104	pre-treatment
Pre-treatment (preferred)	**0.303**	**0.220**	**0.383**	**0.138**	**0.104**	**confounder-only**
Sentinel indices	0.157	0.051	0.239	0.703	0.083	possible mediator
Sentinel bands	0.121	0.034	0.210	0.463	0.051	possible mediator
Full RS context	0.244	0.126	0.345	0.703	0.061	over-adjusted
PCA context	0.231	0.140	0.324	0.560	0.065	over-adjusted

lowers the building-level estimate to 0.244°C, but remains an over-adjustment comparison.

Residual Moran's I remained significant for the pre-treatment specification ($I = 0.112$, $p = 0.01$) and full-context set ($I = 0.102$, $p = 0.01$). Under the 2026-06-30 evidence-grade rule set, the default threshold $t_{\mathrm{Moran}} = 0.10$ triggers `material_residual_moran` and grades Chongqing as bounded support. At thresholds 0.15 and 0.20, the same residual becomes a significant sub-threshold warning. The claim is therefore an outcome-scale, measured-context association with residual-spatial caution, diagnostic building-level matching, and possible high-rise spillover.

Table 5 collects reviewer-critical boundaries using public aggregate artifacts written to `07_results/chongqing_estimand_sensitivity_audit.csv`. It guards against reading the pixel coefficient as a building ATT, treating the near-threshold match as strictly balanced, or treating the residual-Moran grade as a universal validity class.

Table 5: Chongqing estimand and sensitivity audit generated from public aggregate outputs. The table records the claim boundary associated with each reviewer-critical diagnostic.

Dimension	Quantitative evidence	Claim boundary	Manuscript action
Outcome-scale estimand	Pixel high-rise-share slope = 0.546°C per unit share; building-level matching ATT = 0.303°C	Primary result is an outcome-scale association/slope, not a building-level ATT	Rename the primary effect and keep matching as diagnostic
Pixel-cluster precision	Naive SE = 0.038; pixel-cluster SE = 0.054; SE inflation = 41.5%	Building-level precision is overstated unless shared outcome pixels are acknowledged	Report clustered and pixel rows together
Balance near miss	Max post-match SMD = 0.104; balance pass = false	The preferred design is near-threshold, not strictly balanced	Keep the match under bounded support
Residual threshold margin	Moran's $I = 0.112$; margin above 0.10 = 0.012; grade flips at 0.15 and 0.20	Grade is threshold-sensitive, while the residual warning remains	Report the threshold-margin audit
Process transfer	TPR at $t = 0.10$: CAR 0.194, SAR 0.529, kernel 0.994, non-stationary 1.000	Residual Moran's I is process-dependent, not a universal bias detector	State transfer limits wherever the rule is interpreted
Sentinel role boundary	Full-RS max SMD = 0.061 and ATT = 0.244°C, but surfaces may be post-treatment	Better balance does not make Sentinel-rich adjustment causally primary	Keep Sentinel as over-adjustment comparison
Restricted data	Raw Chongqing inputs are not redistributed; public reconstruction is structural	Exact external reproduction needs approved local inputs	State aggregate-audit scope in Data Availability

The variable-role audit (Table 6, source `07_results/chongqing_variable_role_audit.csv`) separates admissibility from balance. Coordinates and terrain are low-risk context variables, geometry is a medium-risk morphology proxy, and Sentinel-derived features are ambiguous proxies or possible mediators. The preferred set trades slightly weaker balance for a more defensible causal role.

4.5 County ArcGIS-free GIS and spatial-diagnostic validation

The county run tests whether the shared SCCA core can reproduce a familiar GIS-facing county analysis in an open notebook workflow, write portable GIS outputs, and show whether spatial diagnostics change the interpretation. The exposure is county social association, the outcome is average age at death, and the unit is the contiguous U.S. county.

Table 6: Chongqing variable-role audit for candidate spatial-context sources. Risk describes possible post-treatment or mediator contamination, not statistical balance.

Context group	Causal role	Risk	Variant	ATT (°C)	Max post SMD	Balance pass
Coordinates	Spatial proxy confounder	Low	coordinates only	0.267	0.080	Yes
Building geometry	Proxy confounder or pre-treatment morphology	Medium	geometry	0.252	0.125	No
Terrain	Pre-treatment confounder	Low	terrain	0.303	0.104	No
Pre-treatment (preferred)	Confounder-only adjustment set	Low	pre_treatment	0.303	0.104	No
Sentinel indices	Ambiguous proxy or mediator	High	sentinel indices	0.157	0.083	Yes
Sentinel bands	Ambiguous proxy or mediator	Medium	sentinel bands	0.121	0.051	Yes
Full RS context	Over-adjusted (possible mediators)	Medium	full RS context	0.244	0.061	Yes

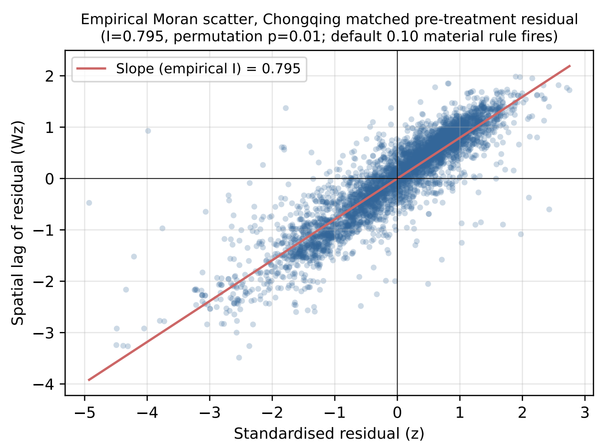

Figure 5: Empirical Moran scatter for the Chongqing matched pre-treatment residual (standardised residual on the x-axis, its spatial lag on the y-axis). The slope is the empirical Moran's $I = 0.112$, statistically significant under 99 permutations ($p = 0.01$); under the default 0.10 rule it triggers a material residual-spatial caution, while under a high-specificity 0.20 sensitivity threshold it would be recorded as significant-below-material.

After 1%–99% exposure trimming, 3,044 of 3,108 counties remained, matching the ArcGIS example count. Baseline adjusted OLS estimated 0.181, while spatial-lag sensitivity estimated 0.145. Residual Moran's I remained 0.313 ($p = 0.010$), and neighbor exposure was significant ($p < 0.001$). The rule table therefore assigns bounded support: the positive association is useful operationally, but not a strong causal claim.

The county run is therefore a reproducibility check, not independent identification evidence or a claim to reproduce proprietary ArcGIS internals.

4.6 Core-support positive control

Because the empirical cases are bounded, we include a positive control (output `07_results/core_support_positive_control.json`). It generates a clean DGP with $n = 3{,}000$, measured confounders, excellent overlap, and no latent spatial process. The control recovers the true effect (ATT = 0.505 vs. true 0.5; relative bias < 1%), passes balance, has non-significant residual Moran's I and neighbor exposure, and receives *core support*. The bounded empirical grades therefore reflect diagnostics rather than a rule engine that cannot issue core support.

4.7 Baseline-to-SCCA comparison

The direct comparisons are stored in `07_results/scca_method_comparison.csv`. In Chongqing, the raw difference was 0.238°C, the preferred pre-treatment estimate was 0.303°C, and the over-adjusted full set gave 0.244°C. In the county case, the spatial layer reduced the coefficient from 0.181 to 0.145 and downgraded evidence because residual Moran's I and neighbor exposure remained significant.

4.8 Threshold calibration for the residual-Moran rule

We calibrated the material residual-Moran threshold t_{Moran} across SAR, CAR, distance-kernel, and non-stationary latent processes. The script sweeps $\rho \in \{0.2, 0.5, 0.8\}$, $n \in \{400, 900, 1600\}$ and $\sigma_u \in \{0, 0.5, 1, 2\}$ for 20 replicates per cell and family, producing 2,880 SCCA fits with known ATT = 0.5. For each fit it records material bias (relative bias > 25%) and threshold firing; outputs are stored in `07_results/`.

Table 7 reports the pooled trade-off and Figure 6 the ROC. Pooling across families, the Youden's J optimum sits at $t = 0.05$ (TPR = 0.79, FPR = 0.03), while $t = 0.10$ keeps a low false-positive rate (0.02) and recovers 68% of materially biased fits. The 2026-06-30 rule set uses $t_{\text{Moran}} = 0.10$ as the default conservative threshold and treats 0.20 as a high-specificity sensitivity threshold.

The trade-off is *family-dependent* (Table 8). Kernel and non-stationary fields produce large biased residual Moran's I (means 0.32 and 0.40), with TPR $\geq$ 0.99 even at $t = 0.20$. CAR fields leave small residual Moran's I (mean 0.04), so $t = 0.10$ recovers only 19% of biased CAR fits and $t = 0.20$ essentially none.

SAR is intermediate (TPR 0.53 at $t = 0.10$). The rule is therefore a useful but process-dependent warning boundary, not a universal detector.

Table 7: Pooled calibration of the SCCA residual-Moran downgrade rule across four spatial-process families (2,880 fits). Positive class: relative bias > 25%. Predictor: $|I_{\text{residual}}| \geq t$ AND permutation $p \leq 0.05$.

Threshold t	TPR	FPR	Precision	Youden's J	Verdict
0.05	0.79	0.030	0.99	0.762	pooled Youden optimum
0.10	0.68	0.023	0.99	0.657	manuscript default; flags Chongqing
0.15	0.59	0.019	0.99	0.573	
0.20	0.57	0.018	0.99	0.556	high-specificity sensitivity
0.25	0.55	0.014	0.99	0.534	
0.30	0.50	0.008	0.99	0.494	too lenient

Table 8: Per-family behaviour of the residual-Moran rule. Mean residual Moran's I is averaged over all cells of that family; TPR is at the manuscript default $t = 0.10$. The rule is strong for kernel/non-stationary processes but weak for CAR, where the same relative bias leaves little simple-contiguity autocorrelation in the residual.

Latent-process family	Mean residual I	TPR @ 0.10	TPR @ 0.20	Family Youden-optimal t
CAR (conditional AR)	0.04	0.19	0.00	0.05
SAR (simultaneous AR)	0.10	0.53	0.30	0.05
Exponential kernel	0.32	0.99	0.99	0.20
Non-stationary	0.40	1.00	1.00	0.05

4.9 Grade-rule sensitivity for the two case studies

The county grade is stable, but the Chongqing grade is boundary-dependent. Varying only the residual-Moran threshold, Chongqing changes from bounded support at 0.10 to a significant sub-threshold warning at 0.15 and 0.20. County remains bounded because residual Moran's $I = 0.313$ and neighboring exposure stay significant.

County response-shape checks also varied the social-capital knot. A knot at 13 gave an after-knot slope increase of 0.104 years per unit (HC3 $p = 0.00075$; AIC improvement = 17.46), but other knots changed strength. This supports the county run as operational sensitivity, not independent causal evidence.

4.10 GIS and notebook outputs

The county notebook also tested SCCA as an operational GIS workflow. The shared core wrote joined tables, GeoPackage, GeoJSON, Shapefile, maps, and

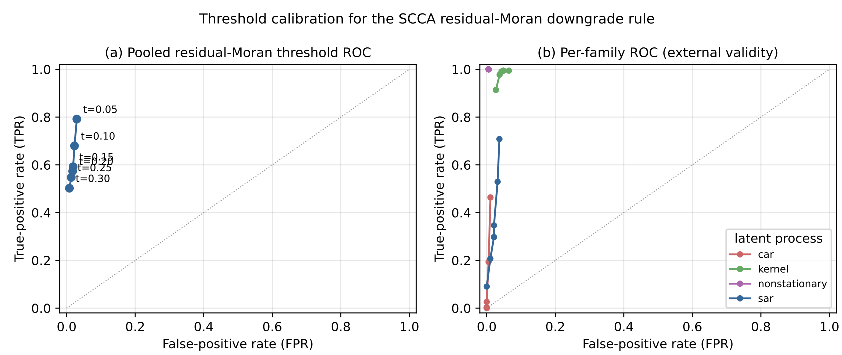

Figure 6: Threshold calibration for the SCCA residual-Moran downgrade rule. (a) Pooled ROC across candidate material thresholds. (b) Per-family ROC: the kernel and non-stationary families sit in the high-sensitivity corner, SAR is intermediate, and CAR is close to the chance-limited corner, showing that the rule's power depends on the unknown latent spatial process.

Table 9: Residual-spatial threshold sensitivity generated by the evidence-synthesis script. The exercise varies only the material residual Moran warning threshold and leaves all other downgrade rules unchanged.

Case	Residual Moran's I	Threshold	Grade	Residual status	Grade rules or flags
Chongqing pre-treatment	0.112	0.10	Bounded	material significant	`material_residual_moran`
Chongqing pre-treatment	0.112	0.15	Core	significant below material threshold	non-downgrade residual warning
Chongqing pre-treatment	0.112	0.20	Core	significant below material threshold	non-downgrade residual warning
County spatial notebook	0.313	0.10	Bounded	material significant	`material_residual_moran`; `significant_neighbor_exposure`
County spatial notebook	0.313	0.15	Bounded	material significant	`material_residual_moran`; `significant_neighbor_exposure`
County spatial notebook	0.313	0.20	Bounded	material significant	`material_residual_moran`; `significant_neighbor_exposure`

QGIS `.qml` styles from the same `AnalysisRequest`. These outputs do not strengthen identification; they make assumptions and spatial cautions visible in GIS environments.

5 Discussion

5.1 What SCCA solves

SCCA addresses a practical gap in geographic causal analysis: spatial variables are often added without showing whether they improved adjustment, and spatial residuals are treated as model diagnostics rather than claim boundaries. SCCA makes the adjustment set inspectable by recording context variants, balance, support, spatial checks, and evidence grades.

The Chongqing case shows why this record matters. Because the outcome is measured on MODIS pixels, the most defensible quantity is the outcome-scale high-rise-share slope, not the finest-scale building match. The paper reports that slope together with diagnostic matching, variable-role and estimand audits, significant residual Moran's $I = 0.112$ ($p = 0.01$), and a bounded grade. The best-balanced Sentinel set attenuates the building-level estimate from 0.303 to 0.244°C, but may be post-treatment. SCCA's contribution is the diagnosed record showing which number has the correct support and where the claim stops.

The county notebook shows the same logic for GIS use. It is not a second substantive validation case; it shows that spatial diagnostics can keep a positive coefficient visible while downgrading support when residual autocorrelation and neighboring exposure remain.

5.2 Practical value for GIS workflows

SCCA converts a causal analysis into an inspectable spatial evidence package. The ArcGIS toolbox, QGIS provider, notebook workflow, and command-line interface share the same request object and manifest, reducing silent divergence across notebooks, desktop joins, and publication tables.

The county comparison gives a concrete example: SCCA reproduced the positive exposure-response direction and sample-trimming count of a commercial GIS analysis, then added robustness, knot-sensitivity, and spatial-diagnostic outputs. This is workflow reproducibility, not causal validation.

GIS outputs are audit objects, not causal evidence by themselves. They keep the evidence boundary attached to the spatial objects used in analysis.

5.3 How SCCA can be extended

The workflow can be strengthened by adding formal sensitivity analysis for unmeasured spatial confounding, making variable-role declarations mandatory, using spatial cross-fitting for high-dimensional nuisance models, and separating SLX-style diagnostics from formal interference estimands [Chernozhukov et al., 2018, Forastiere et al., 2021, Papadogeorgou et al., 2022]. Scale remains central: buffer radius, aggregation, and neighborhood definitions should be part of the adjustment design. The new aggregate Chongqing audit is still not a substitute for latent-confounder sensitivity analysis on restricted building-level data.

5.4 What SCCA does not solve

SCCA does not make observational spatial data equivalent to randomised experiments. It cannot adjust for absent confounders or automatically solve interference. SLX-style summaries are sensitivity outputs, not identified network-interference estimands [Forastiere et al., 2021, Papadogeorgou et al., 2022]. In Chongqing, heat from high-rise clusters plausibly affects adjacent pixels, so SUTVA is unlikely to hold exactly. The building-level ATT can conflate own contrast with spillover; the outcome-scale slope reduces support mismatch but does not identify spillover. SCCA also does not remove MAUP or treatment-outcome scale mismatch [Openshaw, 1984, Fotheringham and Wong, 1991]. Context variables are audited by variants rather than selected automatically, missingness handling remains conservative, and only the residual-Moran rule is calibrated here.

The Chongqing interpretation is therefore bounded by five visible limits: residual spatial structure, building-to-pixel support mismatch, possible post-treatment Sentinel ambiguity, residual-Moran threshold sensitivity, and restricted raw inputs. These are properties of the data that SCCA makes visible.

6 Conclusions

We presented Spatial Context Causal Adjustment, an open diagnostic workflow for geographic observational causal studies. SCCA treats spatial context as a candidate adjustment source and evaluates variants through support, balance, robustness, spatial diagnostics, variable-role audits, and reproducible evidence-grade rules. A 2,880-fit calibration showed that the residual-Moran downgrade is useful but process-dependent. Across synthetic stress tests, Chongqing, and a county notebook/GIS check, SCCA distinguished useful estimates from bounded and fragile designs.

The main empirical result is deliberately bounded. In Chongqing, the outcome-scale pixel model estimated a positive high-rise-share slope for summer LST (0.546°C per unit share, 95% CI [0.359, 0.734]); building-level matching (0.303°C) is diagnostic because treatment is finer than MODIS outcome support. The preferred set slightly missed balance (maximum SMD = 0.104), residual Moran's I = 0.112 crossed the default rule by 0.012, and Sentinel surfaces were treated as possible post-treatment variables. The synthetic benchmark exposed estimator fragility, the positive control was certified as core support, and the county run showed ArcGIS-free reproducibility with spatial downgrading. SCCA's value is not that it removes causal assumptions, but that it makes assumptions, diagnostics, and claim boundaries visible in GIS-facing workflows.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

The repository contains manuscript source, experiment scripts, generated result tables, diagnostic reports, synthetic generators, and redistributable public/example data. It supports exact reruns for the synthetic and county/example analyses and structural Chongqing reruns from approved local inputs. Raw Chongqing geospatial inputs and the building-level UHI sample are not publicly redistributed because they include precise geometries or coordinates, floor attributes, and derived environmental variables. For review, the repository includes non-sensitive aggregate Chongqing audit files with matching settings, estimates, confidence intervals, balance metrics, residual Moran diagnostics, matching sensitivity, threshold-placebo summaries, and the estimand/sensitivity audit. The public repository therefore does not claim full public reconstruction of the restricted Chongqing result.

The county social-capital case uses third-party Esri training/demo data rather than author-generated data. Its metadata credit Esri; the U.S. Census Bureau; NOAA/NOS/NGS; CDC WONDER/NCHS; and 2019 County Health Rankings/Living Atlas data from the University of Wisconsin Population Health Institute and the Robert Wood Johnson Foundation. Use is governed by the Esri Master License and restricted to training, demonstration, and educational purposes. Revised evidence synthesis, rule tables, variable-role audit, estimand/sensitivity audit, method comparison, and public result summaries are stored in `07_results/`.

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